COGNITIVE PERISHABLE INVENTORY INTELLIGENCE PLATFORM FOR AUTONOMOUS EXPIRY RISK PREDICTION AND REAL-TIME DECISION SUPPORT

(P.I.R.E)

A MINI PROJECT REPORT

Submitted by

RESHMA BANU T 312323205182

SAHANA M 312323205187

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

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(An Autonomous Institution)

St. Joseph's Group of

Institution

OMR, Chennai 600 119

ANNA UNIVERSITY:: CHENNAI 600 025

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ANNA UNIVERSITY :: CHENNAI 600 025

BONAFIDE CERTIFICATE

Certified that this Mini project report **"PERISHABLE INVENTORY RISK ENGINE (P.I.R.E)"** is the Bonafide work of **RESHMA BANU T (312323205182)** and **SAHANA M (312323205187)** who carried out the project under my supervision, for the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology.

SIGNATURE

SUPERVISOR

M. Nivethitha Devi M.E., (Ph.D.)
Assistant Professor,
Department of Information Technology,
St. Joseph's College of Engineering,
OMR, Chennai- 600119.

SIGNATURE

HEAD OF THE DEPARTMENT

Dr. C Heltin Genitha, M.E.,Ph.D.,
Professor,
Department of Information Technology,
St. Joseph's College of Engineering,
OMR, Chennai- 600119.

CERTIFICATE OF EVALUATION

College name : St. Joseph's College of Engineering

Branch : Information Technology

Semester : VI

SL.NO	NAME OF THE STUDENTS	TITLE OF THE PROJECT	NAME OF THE SUPERVISOR
1. 2.	RESHMA BANU T (312323205182) SAHANA M (312323205187)	PERISHABLE INVENTORY RISK ENGINE (P.I.R.E)	M. Nivethitha Devi M.E., (Ph.D.)

The report of the Mini Project work submitted by the above students in partial fulfillment for the award of Bachelor of Technology Degree in Information Technology of Anna University were evaluated and confirmed to be reports of the work done by the above students.

Submitted for Mini project and Viva Examination held on________.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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LIST OF ABBREVIATIONS

ABBREVIATION	DEFINITION
IBTrACS	International Best Tracks Archives for Climate Stewardship
RMSE	Root Mean Squared Error
LSTM	Long Short-Term Memory
YOLO	You Only Look Once
API	Application Programing Interface
CNN	Convolutional Neural Network
CSV	Comma- Separated Values
GUI	Graphical User Interface
MAE	Mean Absolute Error
MSE	Mean Squared Error

ABSTRACT

Perishable goods such as food products, pharmaceuticals, and dairy items are highly sensitive to shelf life, storage conditions, and demand variability, making their management a significant challenge in modern supply chains. Inefficient inventory handling often results in product wastage, financial losses, and reduced operational efficiency. Accurate prediction of expiry risk and timely intervention are therefore essential to support sustainability, profitability, and consumer safety.

This study presents a Cognitive Perishable Inventory Intelligence Platform for Autonomous Expiry Risk Prediction and Real-Time Decision Support. The proposed system employs machine learning techniques, including time-series analysis and the XGBoost algorithm, to analyze factors such as remaining shelf life, storage conditions, and consumption patterns. Based on these inputs, the model generates a dynamic risk score and classifies each inventory item into low, medium, or high-risk categories. The system also provides automated alerts and actionable recommendations, including early utilization, discounting, and stock redistribution.

Experimental results demonstrate that the proposed approach effectively identifies high-risk inventory items with high prediction accuracy and low error rates. By enabling proactive inventory monitoring and intelligent decision-making, the system significantly reduces wastage and improves operational efficiency. Its scalable and adaptable design makes it suitable for deployment in retail, healthcare, and logistics environments, supporting sustainable inventory management and optimized resource utilization.

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JUSTIFICATION FOR SDG GOALS

Goal 9 – Build Resilient Infrastructure, Promote Inclusive and Sustainable Industrialization, and Foster Innovation

The project leverages advanced technologies including artificial intelligence and machine learning (time-series models and XGBoost) to transform traditional inventory systems into intelligent decision-support platforms. By integrating predictive analytics with inventory management, the system builds a resilient and scalable digital infrastructure capable of real-time risk assessment and optimization. Such innovations contribute to the development of smart supply chain systems that enhance efficiency and sustainability. This aligns with Target 9.a by promoting technological advancement and supporting the development of sustainable and innovative infrastructure in modern industries.

Goal 12 – Ensure Sustainable Consumption and Production Patterns

Efficient management of perishable goods is essential to reduce waste and promote sustainable consumption. The proposed Perishable Inventory Risk Engine minimizes wastage by predicting the expiry risk of items and enabling timely decision-making. By identifying high-risk inventory and suggesting corrective actions such as early utilization or redistribution, the system helps reduce unnecessary disposal of food and medical products. This directly contributes to sustainable resource utilization and waste reduction. In line with Target 12.3 of the UN SDGs, which aims to halve global food waste, this project strengthens efforts toward responsible consumption and production practices.

CHAPTER 1

INTRODUCTION

1.1 PROJECT OVERVIEW

Perishable goods such as food products, pharmaceuticals, and dairy items are highly sensitive to time, storage conditions, and demand fluctuations, making their management a critical challenge in modern supply chains. Inefficient inventory handling often leads to wastage, financial losses, and reduced operational efficiency. Ensuring timely utilization of such goods is essential for both economic sustainability and consumer safety.

Traditional inventory systems mainly focus on stock tracking and lack predictive capabilities to identify expiry risks. As a result, organizations face issues like overstocking, understocking, and disposal of expired products. This highlights the need for intelligent systems that can proactively manage perishable inventory.

With advancements in machine learning, it is now possible to develop systems that provide predictive insights for better decision-making. In this project, we propose a Perishable Inventory Risk Engine that analyzes factors such as remaining shelf life, storage conditions, and consumption patterns to predict the risk associated with perishable items. The system uses machine learning techniques, including time-series analysis and ensemble models, to generate accurate risk predictions.

To minimize wastage, the system includes an alert mechanism that identifies high-risk items and suggests actions such as early utilization, discounting, or redistribution. Additionally, a user-friendly web interface allows users to input inventory data and visualize risk levels in real time. This integrated approach supports efficient inventory management, reduces waste, and enables data-driven decision-making.

1.2 OBJECTIVES

1. To develop an intelligent perishable inventory risk prediction model using machine learning techniques such as time-series analysis and ensemble methods (e.g., XGBoost), to accurately estimate the expiry risk of items based on factors like shelf life, storage conditions, and demand patterns. To enable dynamic user input of inventory parameters through a web-based interface.

2. To generate and visualize risk levels of inventory items through an interactive dashboard, enabling efficient monitoring and decision-making. To collect and process inventory data including expiry dates, stock levels, and consumption rates for effective analysis.

3. To implement a risk scoring mechanism that categorizes items into low, medium, and high-risk levels, and to compare inventory status over time to identify patterns of wastage and inefficiency, providing quantitative insights into inventory performance.

4. To integrate the risk prediction, monitoring, and alert modules into a unified, user-friendly web platform that delivers real-time alerts, recommendations, and decision-support for efficient perishable inventory management.

CHAPTER 2
LITERATURE REVIEW

1. World Bank (2022), "Digital Transformation of Small and Medium Enterprises in Developing Economies," ***World Bank Group Report.***

The World Bank (2022) study focuses on how SMEs in developing countries are gradually adopting digital systems for business operations. It identifies inventory management as one of the weakest areas in traditional retail setups. While digital systems improve transparency and efficiency, limited technical awareness and infrastructure continue to restrict widespread adoption.[1]

2. FAO (2022), "Food Loss and Waste in Retail Supply Chains," ***Food and Agriculture Organization Report.***

FAO (2022) reports that a significant amount of food waste occurs at the retail level due to poor inventory tracking and expiry mismanagement. The report emphasizes the need for better monitoring systems in small businesses to reduce financial and environmental losses.[2]

3. OECD (2023), "SME and Entrepreneurship Outlook: Digitalization and Productivity in Small Businesses," ***OECD Publishing.***

The OECD report (2023) highlights the growing importance of digital tools in improving productivity and efficiency in small and medium enterprises. It emphasizes that many SMEs still rely on manual inventory practices, leading to inefficiencies such as stock mismanagement and wastage. The report suggests that digital transformation can significantly improve decision-making and operational control, although cost and adoption barriers remain a challenge for small businesses.[3]

4. Rahman, M. and Islam, M. (2024), "Web-Based Inventory Monitoring Systems for Small Businesses," ***International Journal of Information Management Data Insights.***

Rahman and Islam (2024) propose web-based inventory monitoring solutions for SMEs. The study highlights improved visibility of stock levels and better operational control. However, it also identifies issues such as inconsistent data entry and lack of integration with real-world shop operations.[4]

5. Gupta, R. and Kumar, A. (2024), "Inventory Management Challenges in Small Retail Businesses," ***International Journal of Retail & Distribution Management.***

Gupta and Kumar (2024) examine common challenges faced by small retailers, particularly in maintaining accurate stock records. The study found that manual tracking often leads to overstocking or stockouts, directly affecting profitability. However, the adoption of digital inventory systems remains low due to cost sensitivity and lack of technical expertise.[5]

The literature review highlights that small and medium enterprises face significant challenges in inventory management, primarily due to manual tracking systems, lack of real-time visibility, and poor expiry monitoring. Studies consistently show that digital transformation can improve efficiency, reduce wastage, and enhance decision-making in retail environments. However, barriers such as cost, complexity, and limited technical expertise still slow down adoption. To address these gaps, the proposed system introduces a simple yet effective digital inventory expiry tracking solution designed specifically for SMEs, enabling better stock control, reduced wastage, and improved operational efficiency.

CHAPTER 3
SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing systems used for managing perishable inventory primarily rely on traditional database systems, manual tracking methods, and basic inventory management software. These systems typically record information such as stock levels, expiry dates, and sales transactions. Common practices like First In First Out (FIFO) are followed to ensure that older stock is utilized before newer items.

While these approaches are widely used, they are limited in their ability to predict and manage risks associated with perishable goods. They do not effectively account for dynamic factors such as fluctuating demand, varying storage conditions, and consumption patterns. As a result, businesses often face issues such as overstocking, understocking, and unexpected product expiration.

Similarly, in the domain of inventory monitoring and decision-making, traditional systems depend heavily on manual inspection and periodic reporting to identify items nearing expiry. These methods are time-consuming, labour-intensive, and prone to human errors, especially when handling large volumes of inventory across multiple locations. Moreover, the lack of automation and real-time analytical capabilities further limits the efficiency and responsiveness of inventory management processes.

3.1.1 Limitations of existing system

The current inventory management systems for perishable goods are largely reactive and lack predictive intelligence. These systems rely on predefined rules and manual monitoring, which are insufficient to handle the complex and dynamic nature of perishable inventory. They are unable to accurately anticipate expiry risks or demand fluctuations, leading to inefficient decision-making.

Additionally, traditional systems fail to consider multiple influencing factors simultaneously, such as storage temperature variations, demand trends, and supply chain delays. This limitation reduces the accuracy of inventory planning and increases the chances of product wastage. Furthermore, these systems often depend on static or delayed data updates, which restrict their ability to respond to real-time changes in inventory conditions.

In terms of operational efficiency, manual tracking and basic reporting mechanisms are both time-intensive and error-prone. These approaches do not scale well for large inventories and cannot provide instant insights required for effective management. The absence of automated alert systems delays the identification of high-risk items, resulting in unnecessary disposal of products. Overall, the lack of automation, real-time processing, and intelligent analysis makes existing systems inadequate for modern perishable inventory management.

3.2 PROPOSED SYSTEM

The proposed system introduces an intelligent and automated approach for managing perishable goods through the Perishable Inventory Risk Engine, which leverages machine learning techniques for predictive risk assessment and decision support. The system is designed to analyze key factors such as remaining shelf life, storage conditions, and consumption patterns to accurately estimate the risk associated with each inventory item.

For risk prediction, the system utilizes machine learning models such as time-series analysis and ensemble methods (e.g., XGBoost) to process historical and real-time inventory data. These models identify patterns and trends to generate a dynamic risk score, classifying items into low, medium, and high-risk categories. This enables proactive identification of items that are likely to expire soon.

To enhance decision-making, the system incorporates an automated alert mechanism that notifies users about high-risk items and provides actionable recommendations such as early utilization, discounting, or redistribution. These alerts help reduce wastage and improve overall inventory efficiency.

The system also includes a user-friendly web-based interface where users can input inventory data and monitor risk levels in real time through interactive visualizations. This integration of predictive analytics, automated alerts, and real-time monitoring provides a comprehensive solution for efficient perishable inventory management. The proposed approach enables organizations to transition from traditional reactive systems to intelligent, data-driven platforms, improving sustainability and operational performance

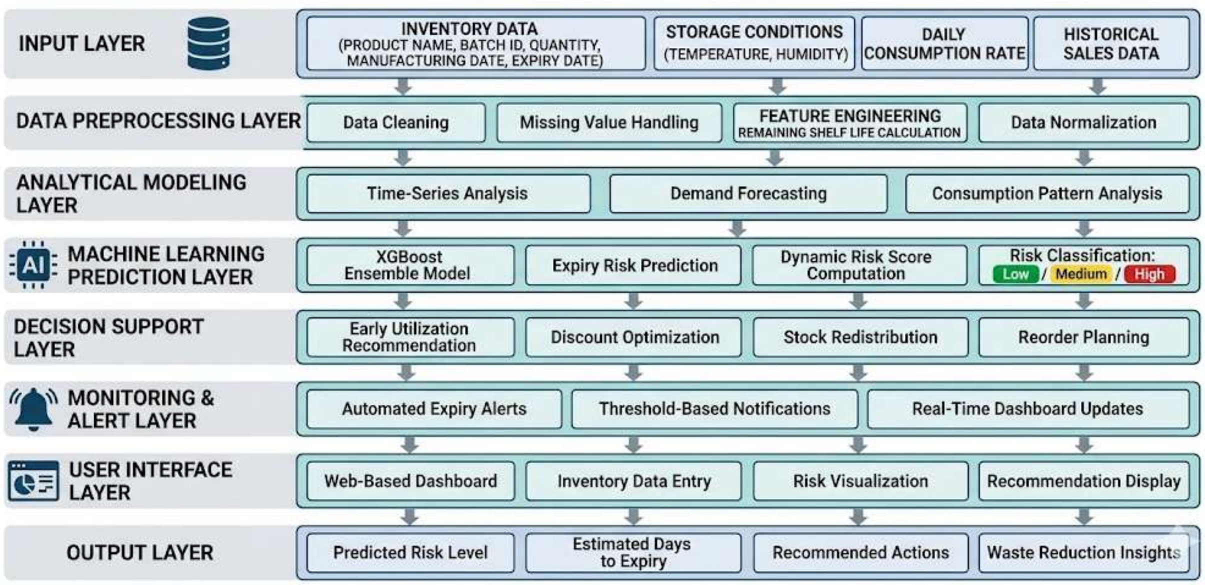


Fig 3.2 Proposed System Diagram

3.2.1 Advantages of proposed system

1. **Enhanced Prediction Accuracy:** Machine learning models such as time-series analysis and XGBoost improve the accuracy of expiry risk prediction by analyzing historical and real-time inventory data

2. **Proactive-Risk Identification:** The system predicts high-risk items in advance, enabling timely action before products expire.

3. **Real-Time Monitoring and Visualization:** A user-friendly web interface allows users to input inventory data and visualize risk levels instantly through interactive dashboards.

4. **Automated Alert System:** The system generates automatic alerts for items nearing expiry, reducing the need for manual monitoring.

5. **Optimized-Inventory Management:** Provides actionable recommendations such as early utilization, discounting, or redistribution to minimize wastage.

6. **Time Efficiency:** Automates inventory analysis and risk prediction, significantly reducing manual effort and processing time.

7. **Scalable Architecture:** The system can be extended to different domains such as retail, healthcare, and logistics with minimal modifications.

8. **Improved Decision-Making:** Supports businesses with data-driven insights, leading to better planning, reduced losses, and efficient resource utilization.

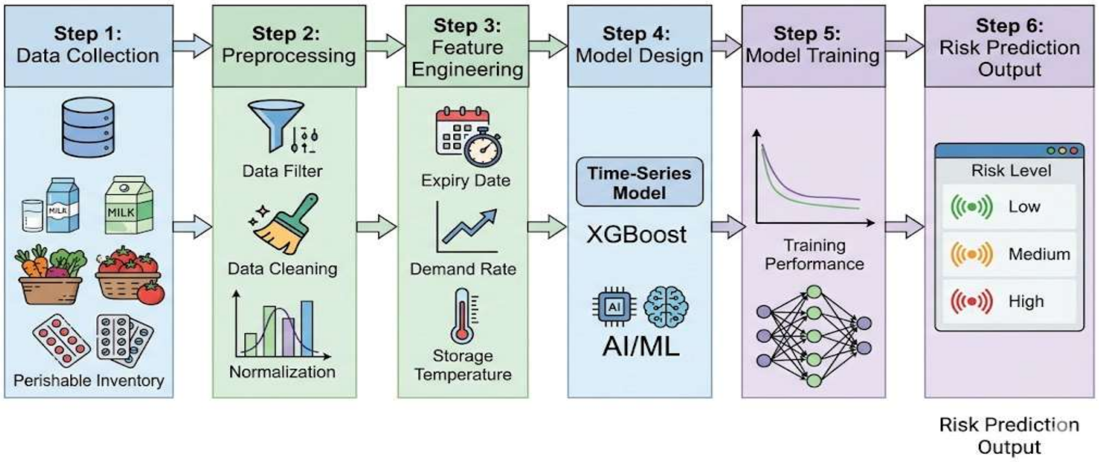
CHAPTER 4
SYSTEM DESIGN

4.1 METHODOLOGY

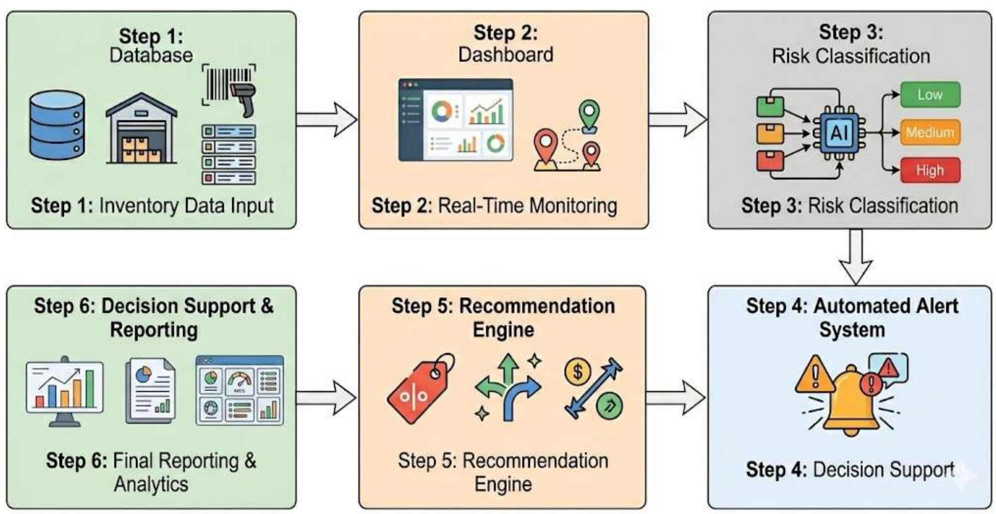
The methodology is divided into two phases, corresponding to the core modules of the system. In the first phase, perishable inventory risk prediction is carried out by applying machine learning techniques such as time-series analysis and ensemble models (e.g., XGBoost) on historical and real-time inventory data. This data includes critical parameters such as expiry date, stock levels, storage conditions, and consumption patterns. The collected data undergoes preprocessing steps including normalization, handling missing values, and feature transformation to ensure compatibility with the machine learning models. Once trained, the model is capable of predicting the risk associated with each inventory item and generating a dynamic risk score based on user-provided inputs through a web interface.

In the second phase, the system evaluates inventory conditions by identifying and analyzing high-risk items based on their predicted risk scores. The system continuously monitors inventory data and categorizes items into low, medium, and high-risk levels. Based on this classification, an automated alert mechanism is triggered to notify users about items nearing expiry. Additionally, the system provides actionable recommendations such as early utilization, discounting, or redistribution of stock to minimize wastage. This approach enables a data-driven assessment of inventory performance and supports proactive decision-making in perishable goods management.

Together, these two modules enable not only the prediction of expiry risk but also the efficient management and optimization of inventory. The integration of predictive modeling with real-time monitoring and alert systems makes the solution both preventive and responsive in nature. Fig 4.1 (a) and (b) represent the methodology of Inventory Risk Prediction and the Inventory Monitoring and Alert System respectively.



(a)



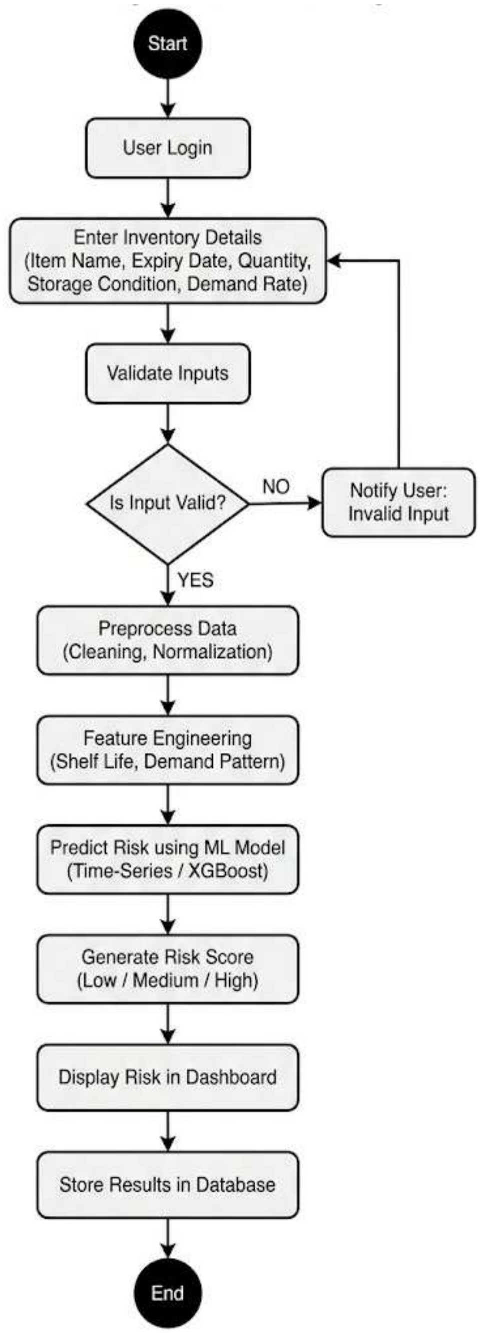
(b)

Fig 4.1 Methodology Diagram (a) Inventory Risk Prediction (b) Inventory Monitoring and Alert System

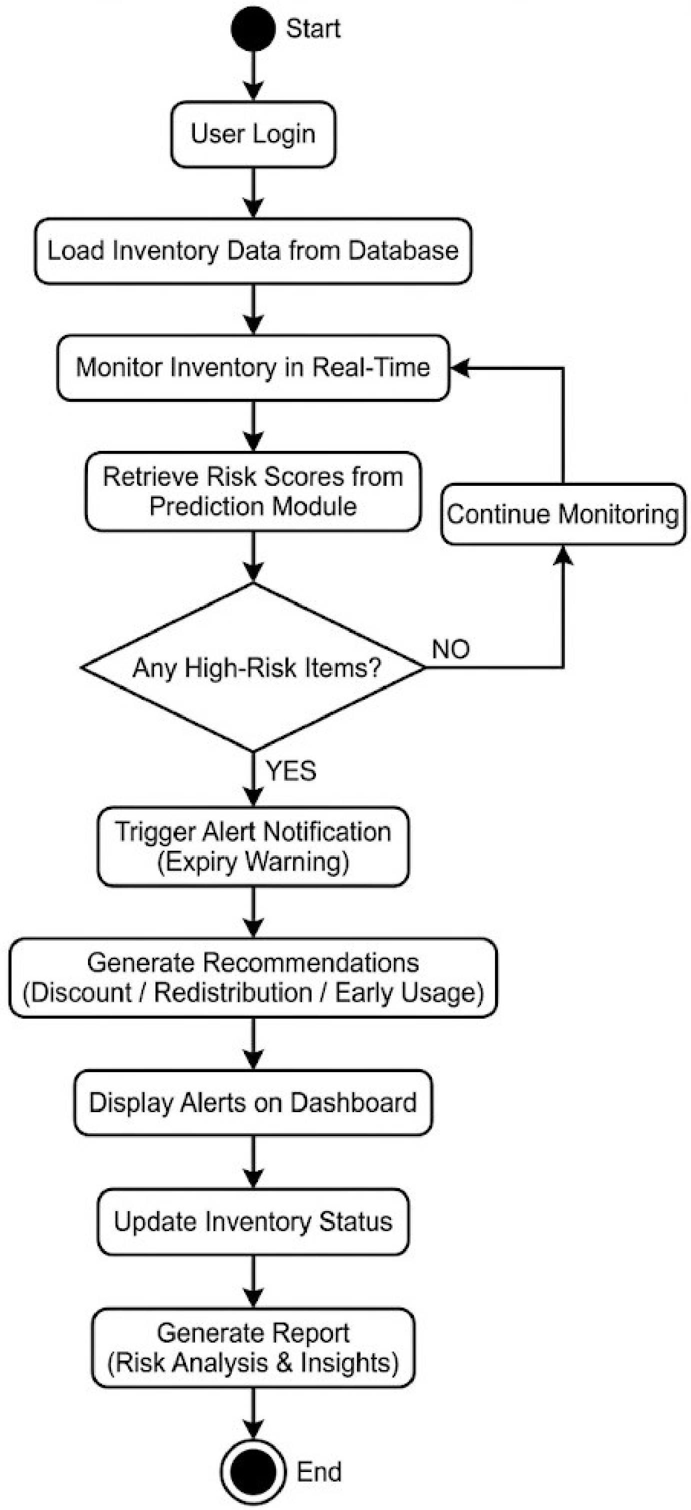
4.2 SYSTEM FLOW

The first stream is responsible for inventory risk prediction. It begins with the collection and preprocessing of historical and real-time inventory data, including parameters such as expiry date, stock levels, storage conditions, and consumption patterns. This data is processed and used to train machine learning models such as time-series models and XGBoost. Once trained, the system can receive live inputs from users or integrated inventory systems. The model uses this input to generate a risk score for each item, categorizing them into low, medium, or high-risk levels. The predicted results are displayed through a dashboard, enabling users to monitor the status of inventory items effectively.

The second stream handles inventory monitoring and alert generation. The processed inventory data is continuously analyzed to track item conditions in real time. Based on the predicted risk levels, the system identifies items that are nearing expiry or are at high risk. An automated alert mechanism is triggered to notify users about such items through notifications or dashboard indicators. Additionally, the system provides recommendations such as early utilization, discounting, or redistribution of stock to reduce wastage. The output is a comprehensive inventory management report that includes risk distribution, wastage reduction insights, and decision-support analytics. Fig 4.2 (a) and (b) represent the system flow of Inventory Risk Prediction and Inventory Monitoring and Alert System respectively.


Start
User Login
Enter Inventory Details (Item Name, Expiry Date, Quantity, Storage Condition, Demand Rate)
Validate Inputs
Is Input Valid?
NO
Notify User: Invalid Input
YES
Preprocess Data (Cleaning, Normalization)
Feature Engineering (Shelf Life, Demand Pattern)
Predict Risk using ML Model (Time-Series / XGBoost)
Generate Risk Score (Low / Medium / High)
Display Risk in Dashboard
Store Results in Database
End

(a)

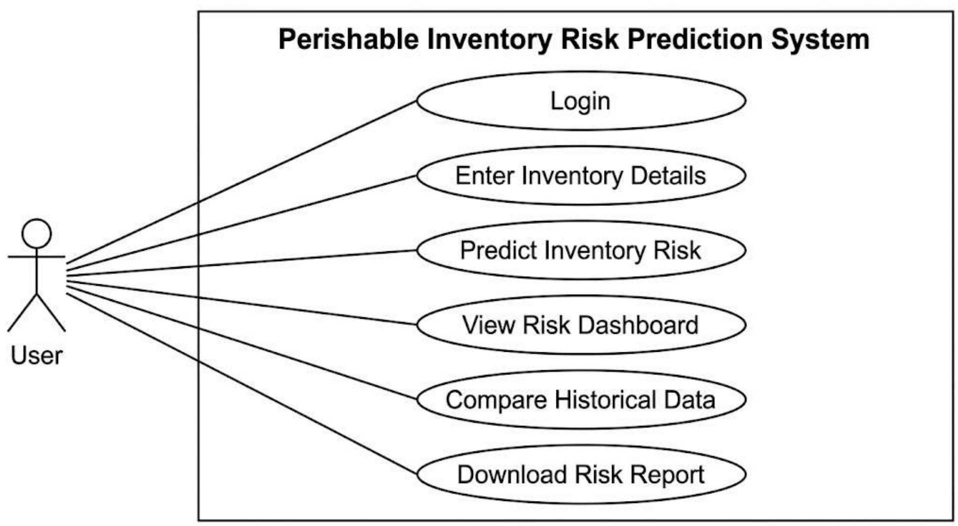


(b)

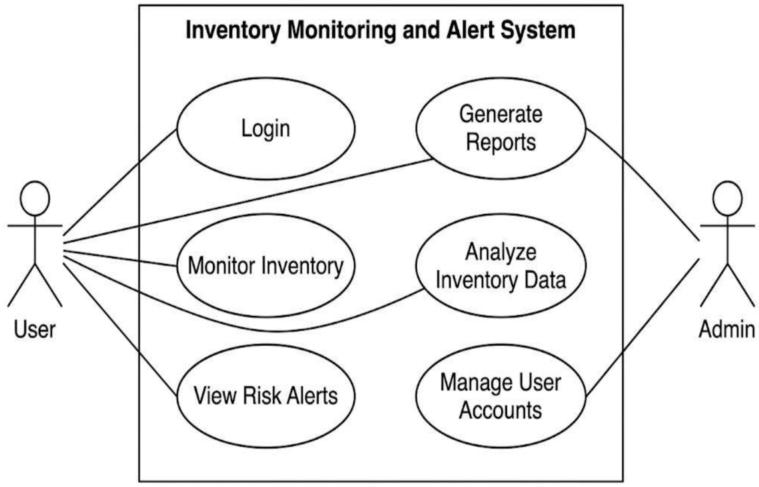
Fig 4.2 System Flow Diagram (a) The system flow of Inventory Risk Prediction (b) Inventory Monitoring and Alert System

4.3 USE-CASE DIAGRAM

The use case for the proposed system involves multiple user roles. The system allows users to input inventory data, preprocess it, and train or load a machine learning model to predict the risk associated with perishable items based on parameters such as expiry date, storage conditions, and demand patterns. Simultaneously, users can monitor inventory status in real time, retrieve predicted risk scores, initiate alert generation for items nearing expiry, and receive an automatically generated report that quantifies inventory risk and potential wastage. The system provides functionalities like viewing, analyzing, and exporting inventory risk data for further decision-making or operational reporting. Fig 4.3 (a) and (b) represents the Use-Case Diagram of Inventory Risk Prediction and Inventory Monitoring and Alert System respectively.



(a)



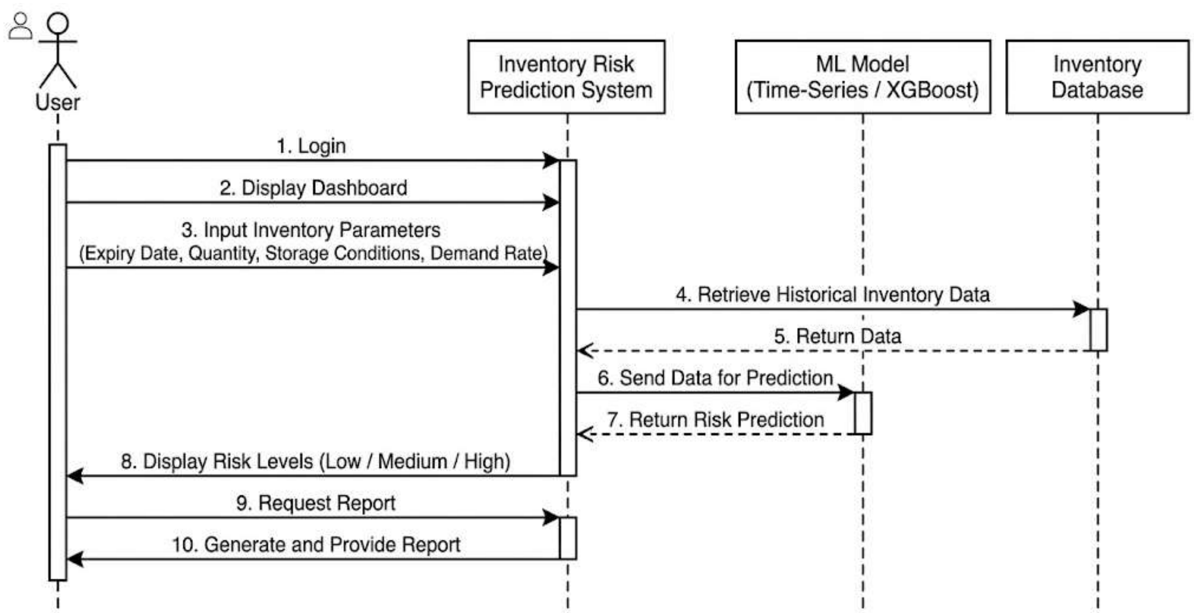
(b)

Fig 4.3 (a) and (b) Inventory Risk Prediction and Inventory Monitoring and Alert System

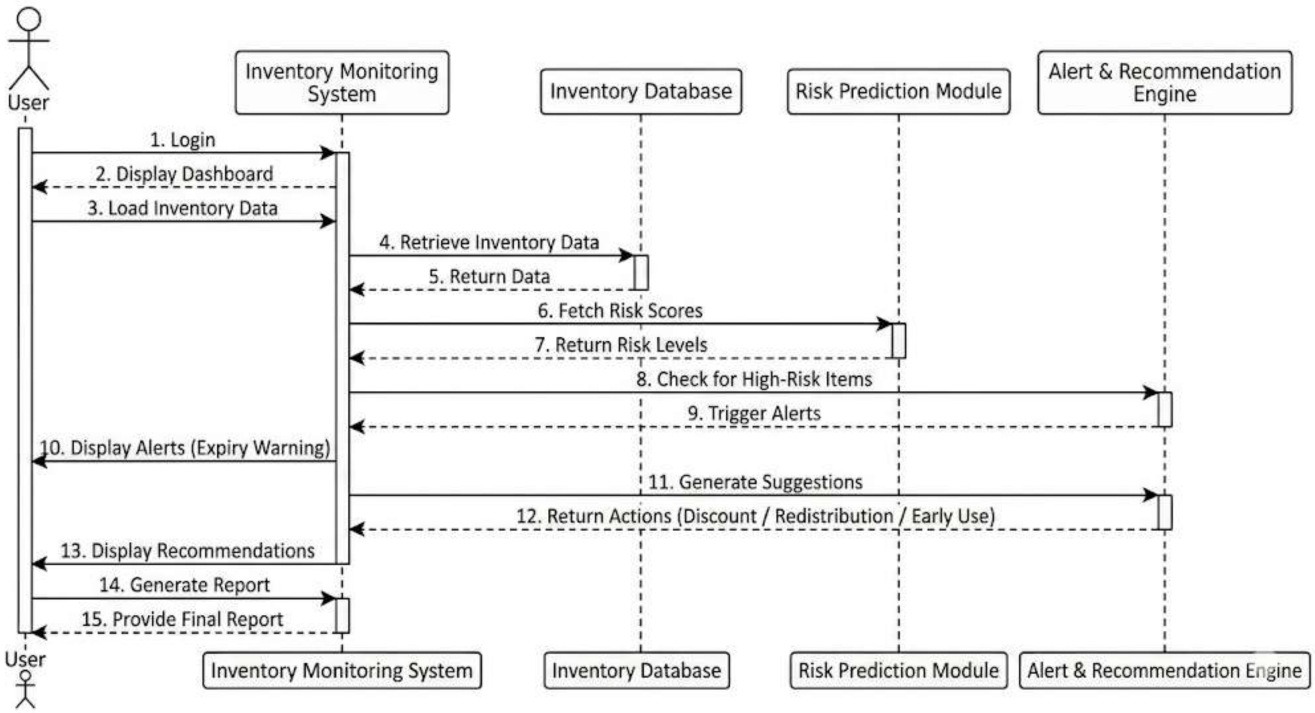
4.4 SEQUENCE DIAGRAM

The combined system consists of two major workflows: inventory risk prediction and inventory monitoring with alert generation. In the inventory risk prediction flow, the process begins when a user inputs inventory details such as item name, expiry date, quantity, storage conditions, and consumption rate into a web interface built using Flask. The values are passed to a backend controller that normalizes and preprocesses the data before feeding it into a trained machine learning model. The model predicts the expiry risk of items based on historical patterns and current conditions. This output is then processed into a risk score and rendered on an interactive dashboard, which is displayed back to the user.

Simultaneously, in the inventory monitoring and alert generation flow, the system continuously tracks inventory data stored in the database. The data is periodically retrieved and analyzed to identify items approaching their expiry threshold. Based on the risk levels generated by the prediction model, the system detects high-risk items and triggers automated alerts. These alerts may include recommendations such as early utilization, discounting, or redistribution. The results are then presented to the user in the form of notifications and visual summaries for effective decision-making. This sequence-based design ensures clear data flow and role separation between the prediction and monitoring modules, allowing parallel execution if necessary. Fig 4.4 (a) and (b) represents the Sequence Diagram of Inventory Risk Prediction and Inventory Monitoring with Alert Generation respectively.



(a)



(b)

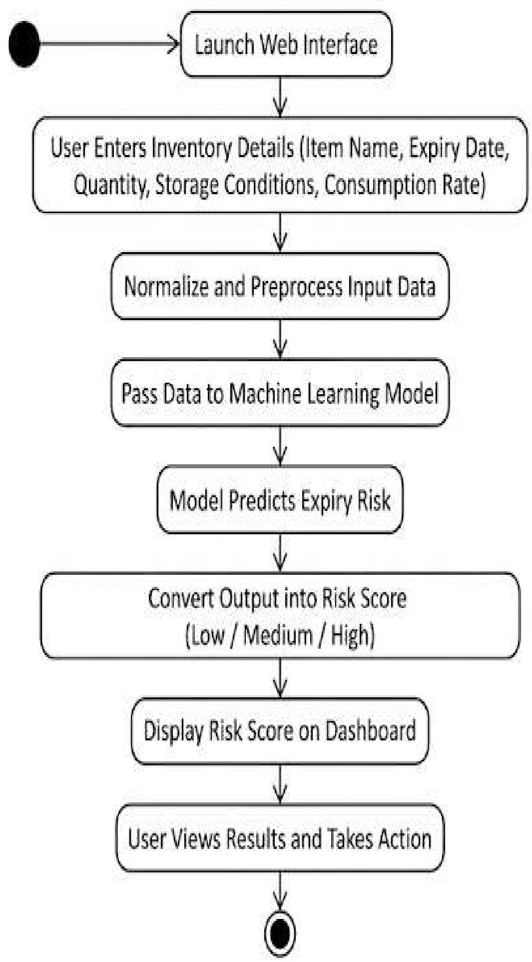
Fig 4.4 Sequence Diagram (a) Inventory Risk Prediction

(b) Inventory Monitoring with Alert Generation

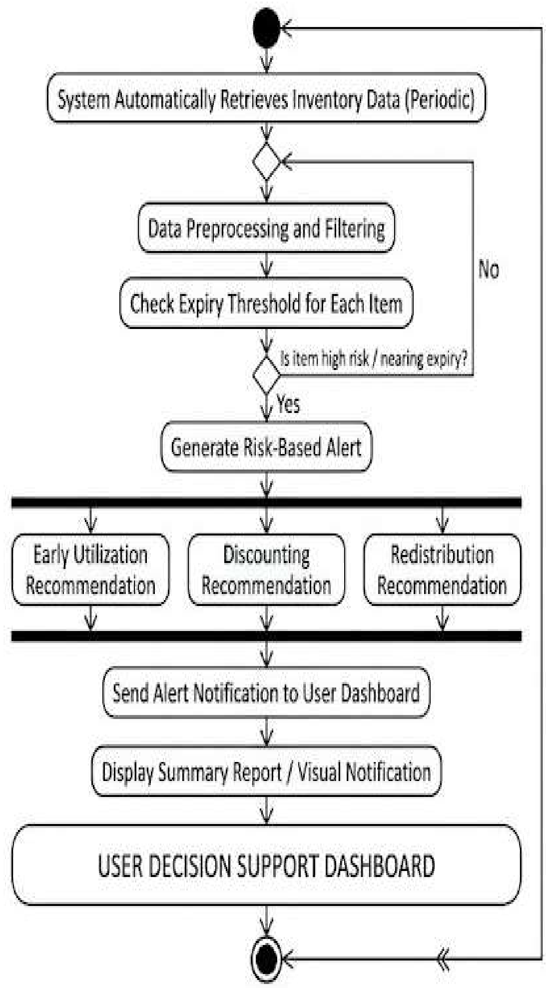
4.5 ACTIVITY DIAGRAM

For inventory risk prediction, the user launches the web interface and enters inventory details such as item name, expiry date, quantity, storage conditions, and consumption rate. These inputs are first normalized and preprocessed before being passed into the machine learning model, which predicts the expiry risk of each item. The result is then converted into a risk score and visualized on a dashboard for easy interpretation. The user can view this information to identify high-risk items and take appropriate actions. In the inventory monitoring and alert generation workflow, the system continuously retrieves inventory data from the database. The data is processed and evaluated to identify items approaching their expiry threshold. Based on the predicted risk levels, the system detects high-risk items and generates automated alerts. These alerts include recommendations such as early utilization, discounting, or redistribution to minimize wastage. A summary report or visual notification is then provided to the user to support effective decision-making.

Each activity culminates in a visual or data-based output designed to improve inventory management and reduce wastage. Fig 4.5 (a) and (b) represents the Activity Diagram of Inventory Risk Prediction and Inventory Monitoring with Alert Generation respectively.



(a)



(b)

Fig 4.5 Activity Diagram (a) the Activity Diagram of Inventory Risk Prediction (b) Inventory Monitoring with Alert Generation

4.6 CLASS DIAGRAM

In the Inventory Risk Prediction system, key classes include UserInputHandler (to collect inventory details such as item name, expiry date, quantity, storage conditions, and consumption rate), DataPreprocessor (for cleaning and normalizing input data), RiskPredictionModel (for predicting expiry risk using machine learning models such as time-series analysis and XGBoost), RiskAnalyzer (to calculate and categorize risk levels), and FlaskController (to manage communication between the user interface and backend processing).

In the **Inventory Monitoring and Alert System**, the main classes are **InventoryDatabaseHandler** (for storing and retrieving inventory data), **ExpiryTracker** (to monitor remaining shelf life of items), **AlertGenerator** (to generate notifications for high-risk items), **RecommendationEngine** (to suggest actions such as early utilization, discounting, or redistribution), and **ResultDisplayer** (to present risk levels, alerts, and summaries visually or textually to the user).

This class structure ensures a modular and organized design that supports both predictive analysis and real-time monitoring, enabling efficient and intelligent management of perishable inventory. Fig 4.6 (a) and (b) represents the Class Diagram of Inventory Risk Prediction and Inventory Monitoring with Alert Generation respectively.

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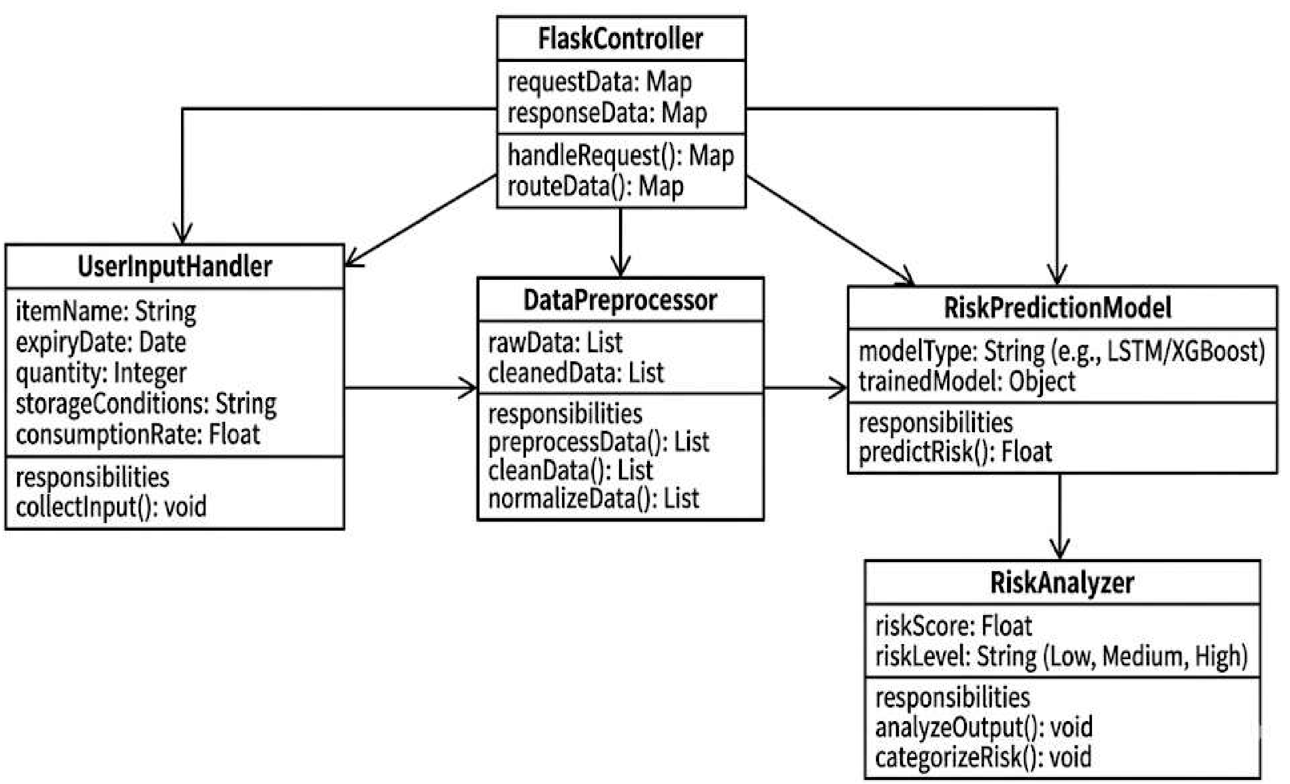
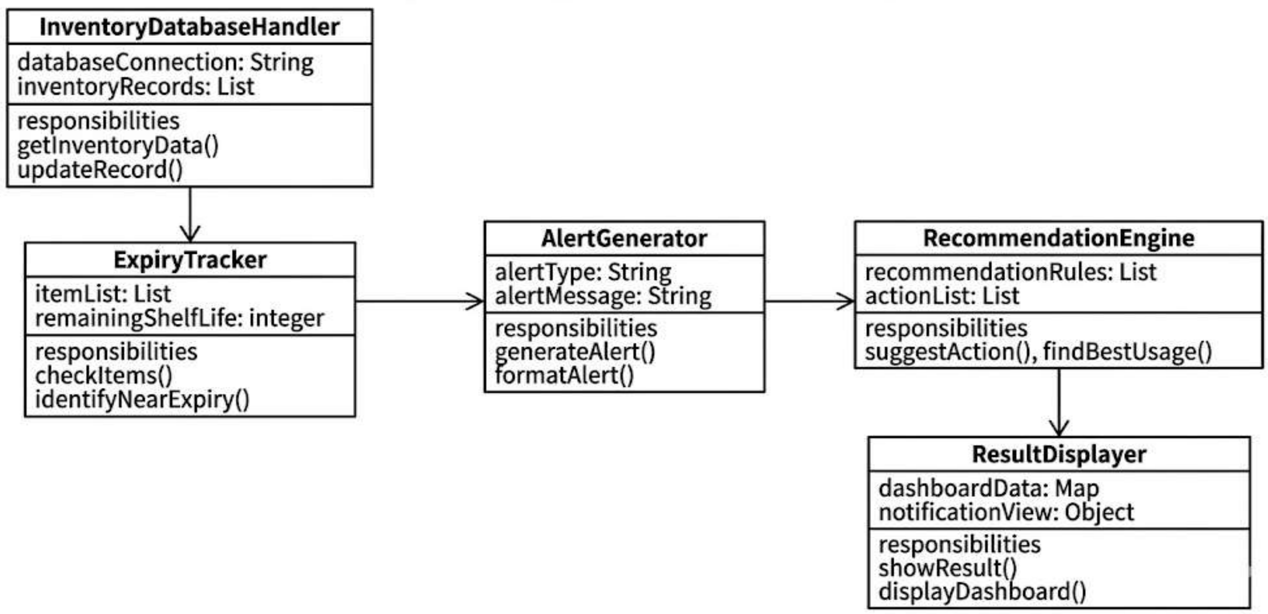


(a)



(b)

Fig 4.6 Class Diagram (a) Inventory Risk Prediction (b) Inventory Monitoring with Alert Generation

CHAPTER 5
SYSTEM IMPLEMENTATION

The implementation of the proposed system involves two core modules: inventory risk prediction using machine learning models such as time-series analysis and XGBoost, and inventory monitoring with automated alert generation for perishable goods. These components were developed and integrated into a web-based interface, allowing users to input inventory details and interact with model predictions and real-time risk analysis seamlessly.

5.1 MODEL DESCRIPTION

The proposed system integrates multiple machine learning models to address two major objectives: inventory risk prediction and inventory monitoring with automated alert generation for perishable items. The first component utilizes time-series analysis combined with XGBoost to forecast the expiry risk of inventory items. These models are designed to process structured inventory data by learning patterns from attributes such as item name, expiry date, quantity, storage conditions, and consumption rate. In the context of inventory forecasting, the model analyzes historical usage and decay patterns to predict the likelihood of an item reaching its expiry threshold. The ability of XGBoost to handle non-linear relationships and feature interactions enables it to effectively capture complex dependencies in inventory behavior. Fig 5.1 represents Model Architecture for Inventory Risk Prediction using Time-Series Analysis and XGBoost.

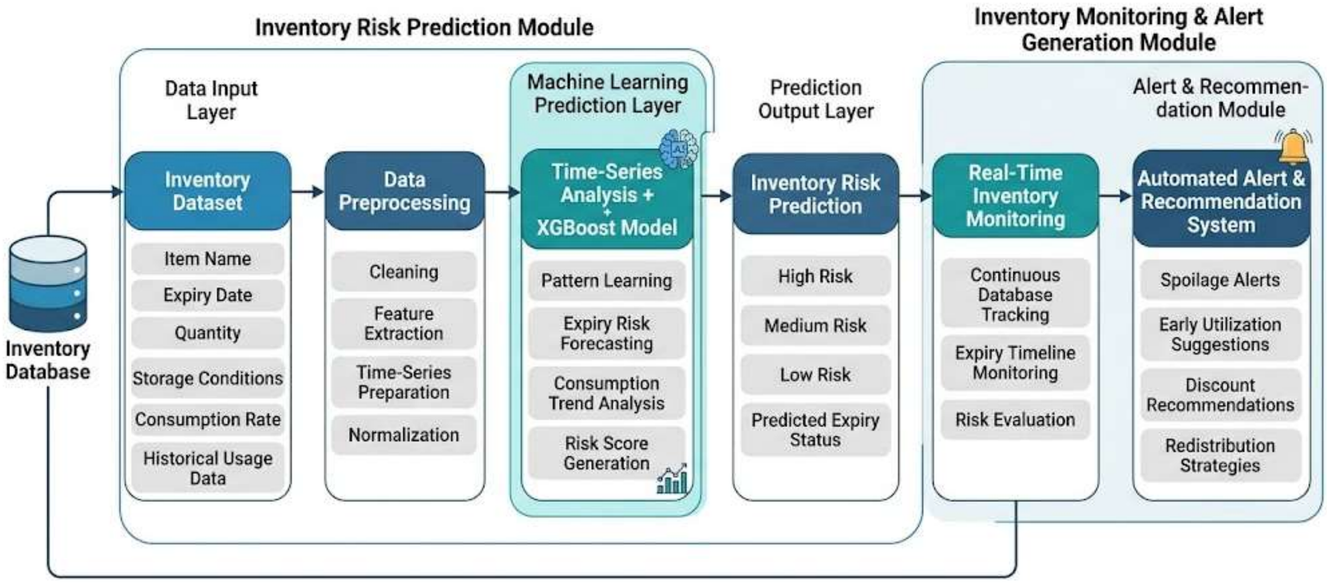


Fig 5.1 Model Architecture for Inventory Risk Prediction using Time-Series Analysis and XGBoost

The second component of the system is the monitoring and alert generation model, which continuously tracks inventory status from the database in real time. This module evaluates expiry timelines and risk scores to identify high-risk items approaching spoilage. Based on the detected risk levels, the system triggers automated alerts along with recommendation strategies such as early utilization, discounting, or redistribution. This monitoring mechanism supports efficient decision-making by ensuring timely intervention and minimizing wastage of perishable goods.

5.2 ARCHITECTURE OVERVIEW

The architecture of the proposed system is composed of two main modules: an inventory risk prediction architecture based on time-series analysis with XGBoost, and an inventory monitoring and alert generation architecture. The prediction module begins with an input layer that receives structured inventory data, including item name, expiry date, quantity, storage conditions, consumption rate, and historical usage patterns. These inputs are processed through a data preprocessing pipeline that handles missing values, normalization, and feature engineering to improve model efficiency. The processed data is then passed through a time-series analysis component that captures temporal consumption trends and usage behavior over time. Following this, feature vectors are forwarded to the XGBoost model, which learns complex relationships between inventory attributes and predicts an expiry risk score. The model is optimized using loss functions that minimize prediction error and improve accuracy during training. Fig 5.2 represents the architecture of the inventory monitoring and alert generation system integrated with the overall prediction framework.

In parallel, the monitoring and alert generation architecture continuously retrieves updated inventory records from the database in real time. This module evaluates expiry timelines and risk scores using a rule-based and threshold-based analysis system to identify items approaching critical expiry levels. The system then triggers an alert generation unit, which produces notifications along with recommended actions such as early utilization, discounting, or redistribution. These alerts are passed to a result display layer that visualizes risk levels, alerts, and summary reports on the user dashboard. This ensures timely decision-making and reduces wastage of perishable goods.

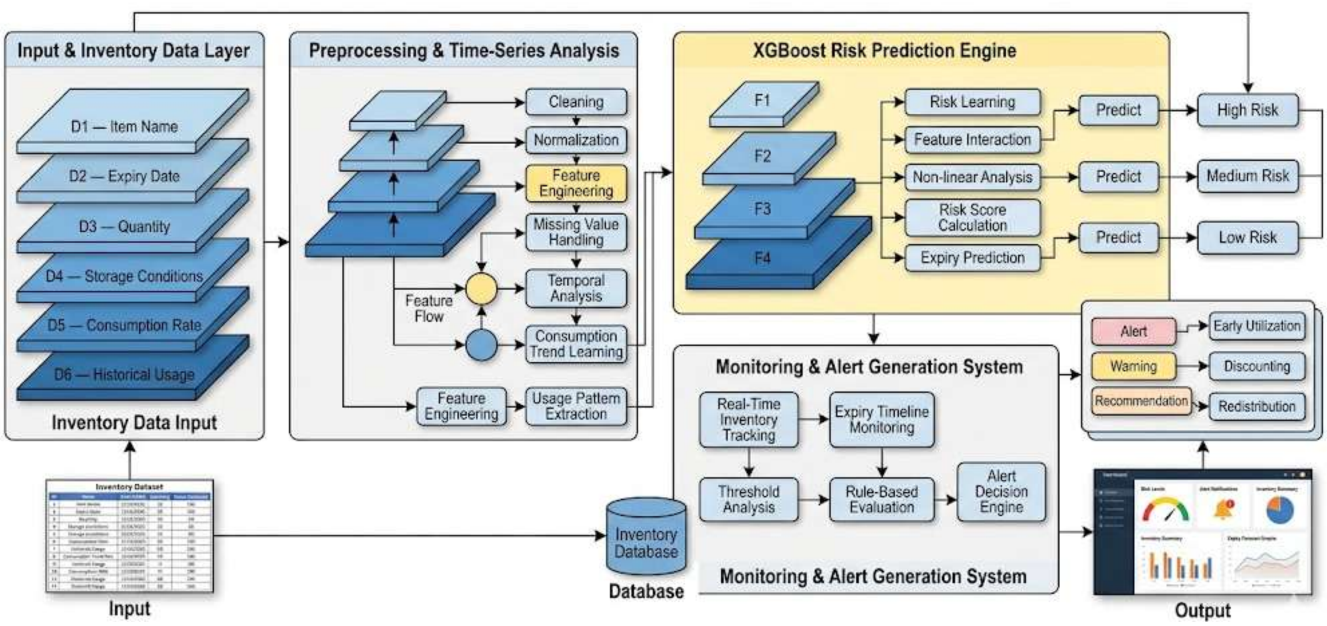


Fig 5.2 Inventory monitoring and alert generation system integrated with the overall prediction framework Architecture Diagram

Together, the XGBoost-based prediction module, time-series analysis component, and real-time monitoring system form a unified and efficient architecture as illustrated in Fig 5.2. While the prediction module focuses on forecasting expiry risk, the monitoring system ensures continuous tracking and immediate alert generation. This integrated design enables intelligent inventory management, improved operational efficiency, and effective reduction of inventory loss.

5.3 METHODOLOGY

This work introduces a machine learning framework designed to predict inventory expiry risk and support real-time monitoring with automated alert generation. The methodology involves data collection, preprocessing, model training, and system evaluation to ensure accurate and reliable results. The entire process is structured to integrate historical inventory records and real-time stock data, enabling a comprehensive analysis of consumption patterns, expiry behavior, and timely identification of high-risk items for effective inventory management.

5.3.1 Data Collection

To build a system that can predict inventory expiry risk and support real-time monitoring with alert generation, we required two types of data: structured inventory data and real-time stock records from nearby shops and storage facilities. For the inventory risk prediction model, we collected datasets from local retail shops, supermarkets, and small businesses, including details such as item name, expiry date, quantity, storage conditions, and daily consumption rate. These features are essential to understand how different products behave in real storage environments and how quickly they approach expiry under varying conditions. Fig 5.3.1 represents the dataset collection process from nearby shops and local vendors used for inventory analysis and risk prediction.

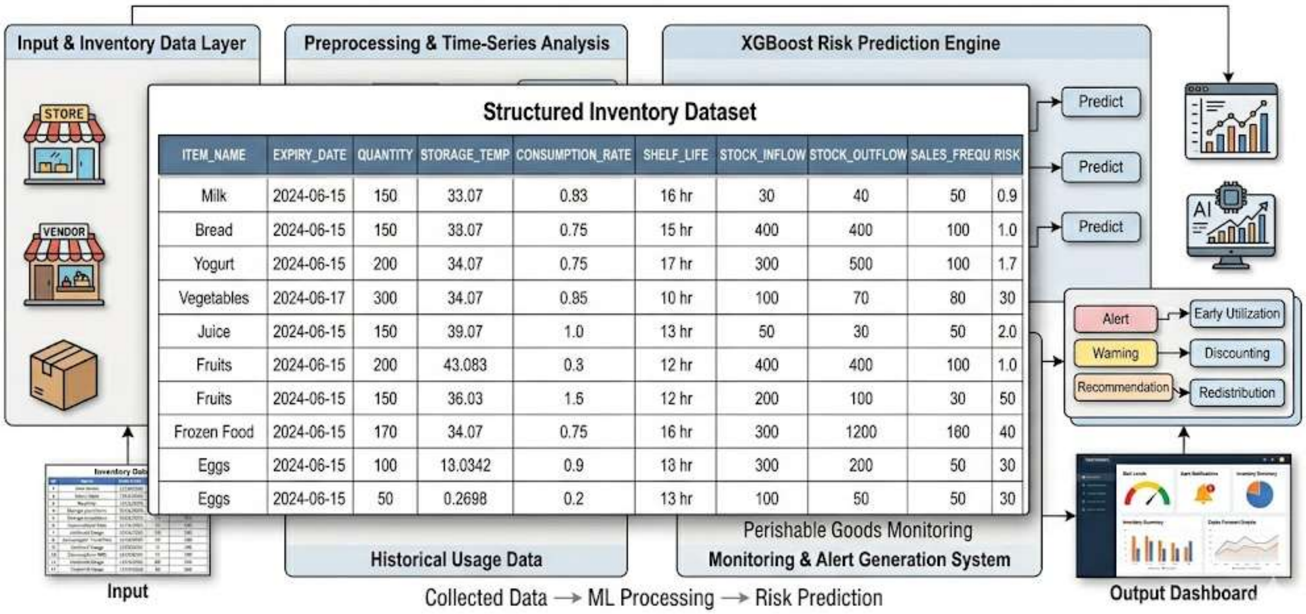


Fig 5.3.1 dataset collection process from nearby shops and local vendors used for inventory analysis and risk prediction

For monitoring and alert generation, we gathered continuous inventory updates from partner shops and vendors, focusing on fast-moving consumer goods and perishable items. This included daily stock inflow and outflow records, sales frequency, and remaining shelf-life information. These real-time records help in tracking product availability and identifying items that are nearing expiration.

Together, these datasets from nearby shops and local inventory sources provide the foundation for a system that not only predicts expiry risk accurately but also ensures timely monitoring and effective reduction of inventory wastage.

5.3.2 Data Preprocessing

Once we gathered the data from nearby shops and local vendors, the next crucial step was to clean and prepare it so our models could learn from it effectively. Starting with the inventory dataset, we ensured that missing values were handled and inconsistencies in stock records were removed. Since machine learning models cannot directly interpret raw categorical and date formats, variables such as item name, storage condition, and expiry dates were converted into numerical representations using encoding techniques. Then, we applied Min-Max normalization to scale important features like quantity, consumption rate, and remaining shelf life into a uniform range. This ensures that no single feature dominates the training process and improves model stability.

After preprocessing, the clean and scaled dataset was split into training and testing sets for model development. From there, we developed prediction models to estimate inventory expiry risk accurately.

On the monitoring side, we processed real-time inventory records to ensure consistency and reliability before analysis. Data from different shops was standardized into a common format and cleaned to remove duplicate or incorrect entries. These preprocessing steps help the system focus on meaningful patterns in stock movement and expiry behavior.

5.3.3 Data Augmentation

To enhance the model's generalizability and improve prediction accuracy, we applied data augmentation techniques on the collected inventory dataset. This helps the system perform better across different shops and real-world variations in stock behavior. For the time-series inventory data, we generated multiple overlapping sequences using a sliding window approach. This allows the model to learn consumption patterns over consecutive time intervals and improves its ability to detect expiry trends more effectively. In addition, synthetic variations were introduced in parameters such as consumption rate and stock movement to simulate real-world fluctuations in demand and supply conditions.

To make the inventory risk prediction model more robust, we also introduced controlled variations in the dataset to handle uncertainties in shop environments. Small perturbations were applied to numerical features like quantity and consumption rate to mimic real-life inconsistencies in reporting and stock updates. This helps the model generalize better across different retail scenarios and reduces overfitting.

These augmentation techniques ensure that the model learns from a more diverse and realistic dataset, improving its stability and reliability in predicting expiry risk across various inventory conditions.

5.3.4 Model Development

For inventory risk prediction, we developed a hybrid machine learning architecture combining time-series analysis and XGBoost, an ensemble learning method. The time-series component was designed to capture temporal dependencies in inventory behavior, such as changes in stock levels, consumption rate, expiry progression, and demand patterns over time. This helps identify trends and fluctuations in product usage across different retail environments.

The time-series model was trained using sequential input data structured based on samples, time steps, and features derived from inventory records. The XGBoost model was integrated to leverage its boosting capability and ability to handle complex feature interactions. This ensemble approach builds multiple decision trees iteratively, where each tree improves upon the errors of the previous ones, resulting in higher prediction accuracy for expiry risk estimation.

To make the system accessible and interactive, we developed a real-time web application using Flask and HTML, allowing users to input inventory details and view risk predictions through a dashboard interface. The user inputs include item name, expiry date, quantity, storage conditions, and consumption rate, which are processed by the trained models to output an expiry risk score.

For inventory monitoring and alert generation, we implemented a rule-based decision system integrated with predictive outputs to track high-risk items in real time. This module continuously evaluates stock records and triggers alerts when items approach critical expiry thresholds. The system plays a key role in identifying potential wastage and supporting timely decision-making in inventory management.

Together, these models form a unified pipeline that predicts inventory expiry risk and enables efficient monitoring with automated alerts, ensuring better stock control and reduced wastage of perishable goods.

5.3.5 Model Training

Once the models were developed, we moved into the training phase where each model was fine-tuned to perform its specific task effectively. The time-series based inventory prediction model was trained using sequential inventory data collected from nearby shops. It consists of structured feature inputs, allowing it to learn long-term consumption and expiry patterns essential for accurate risk prediction. The model was trained using Mean Squared Error (MSE) loss function and optimized using the Adam optimizer with a learning rate of 0.001. Backpropagation was used to update the model weights. The XGBoost model was trained on the same dataset with 100 decision trees, and early stopping was applied to prevent overfitting. The dataset was split into training and testing sets to evaluate model performance on unseen inventory data.

For inventory monitoring and alert generation, rule-based thresholds and predictive outputs were trained to identify high-risk items approaching expiry. The system was tuned to ensure accurate detection of spoilage-prone items across different shop environments.

The model was evaluated using a hold-out validation approach where 20% of the dataset was reserved for testing. This ensures the reliability and generalization of the system across real-world inventory conditions.

5.3.6 Model Evaluation

After training, the models were thoroughly evaluated using multiple performance metrics to ensure they could generalize well to unseen data.

Model Evaluation Metrics: Model Evaluation Metrics:

1. Test Loss (MSE): 0.0041
2. Test MAE (Mean Absolute Error): 0.0428

These metrics indicate that the model is able to predict inventory expiry risk with a small average error (~0.0428), which is reasonable for real-world inventory forecasting in dynamic retail environments.

These evaluation metrics provided a comprehensive picture of prediction accuracy. The results showed that XGBoost outperformed the time-series model, achieving lower MAE and RMSE values, which indicates better performance in minimizing prediction errors. While the time-series model captured sequential patterns in inventory usage effectively, XGBoost's ability to handle complex feature interactions led to more accurate risk predictions overall.

For inventory monitoring and alert generation, performance was evaluated based on the accuracy of high-risk item detection and the timeliness of alert generation. The system's predictions were compared with actual expiry outcomes, and results showed strong alignment with real inventory conditions.

Furthermore, residual error plots and performance visualizations were generated to analyze model behavior. These plots highlighted prediction deviations, while dashboard-based visual outputs provided a clear representation of high-risk inventory zones, helping identify items most likely to expire soo.

5.3.7 Impact Analysis

The combined approach of inventory risk prediction and real-time monitoring with alert generation offers a practical framework for efficient inventory management and wastage reduction. The inventory risk prediction model, accessible via a web application, allows users to input real-time parameters such as item name, expiry date, quantity, storage conditions, and consumption rate. This user-friendly interface helps visualize predicted expiry risk scores on a dashboard, enabling shop owners and managers to make timely stocking, utilization, and procurement decisions.

On the other hand, the inventory monitoring system leverages continuous database updates and rule-based analysis to estimate potential stock loss due to expiry. By calculating the difference between expected usage and remaining shelf life, this method provides a quantifiable measure of inventory risk. Although manual observation can give a general idea of stock status, this data-driven approach delivers concrete and scalable metrics, enabling targeted interventions such as prioritizing early utilization, applying discounts, or redistributing stock to minimize wastage and improve operational efficiency.

CHAPTER 6

CONCLUSION

P.I.R.E work presents a comprehensive machine learning framework for both predicting inventory expiry risk and enabling real-time monitoring with automated alert generation. By integrating time-series analysis with XGBoost, the inventory risk prediction system demonstrates the effectiveness of combining sequential pattern learning with ensemble methods to achieve improved accuracy in forecasting expiry behavior. The accompanying web application empowers users such as shop owners and inventory managers with an intuitive platform to input item details like expiry date, quantity, storage conditions, and consumption rate, and receive visualized risk scores on a dashboard, enhancing timely decision-making and stock control strategies.

The inventory monitoring and alert generation module continuously tracks stock data from the database and identifies high-risk items nearing expiry. By generating automated alerts along with recommendations such as early utilization, discounting, or redistribution, the system quantifies potential inventory loss and helps minimize wastage. This data-driven approach offers more scalable and consistent insights than manual monitoring, supporting efficient inventory management and operational optimization.

Together, these modules form a holistic system capable of supporting both proactive prediction and reactive monitoring, significantly contributing to improved inventory control, reduced wastage of perishable goods, and smarter decision-making in retail environments. these modules form a holistic system capable of supporting both proactive preparedness and reactive impact analysis, significantly contributing to cyclone disaster mitigation.

6.1 FUTURE SCOPE

Several enhancements can be made to expand the utility and robustness of this project:

1. Real-Time Data Integration: In future implementations, real-time inventory data such as stock levels, sales transactions, and expiry updates can be automatically fetched from shop management systems or POS databases. This would eliminate manual entry and allow business owners to continuously monitor inventory status and receive instant expiry risk predictions through the web application.

2. Web Application for Monitoring and Alerts: While the current monitoring module operates with periodic updates, a future version could be extended into a fully interactive real-time web application. This would allow live tracking of inventory movement and immediate alert generation when items approach expiry, benefiting retailers, warehouses, and supply chain managers in decision-making and stock optimization.

3. Expansion to Multiple Product Categories: The framework could be generalized to handle a wider range of products, including dairy, pharmaceuticals, and packaged goods, using transfer learning and improved feature engineering. This would make the system applicable across different industries and retail environments.

4. Geospatial and Store-Level Risk Mapping: Integrating the system with location-based analytics could enable store-wise or region-wise risk visualization, allowing managers to identify high-wastage zones and optimize distribution and inventory planning more effectively.

5. Incorporation of Economic and Demand Data: Integrating additional factors such as market demand, seasonal trends, and product value could improve prediction

accuracy and help prioritize high-value or fast-moving items for better inventory control.

6. By building upon this foundational work, the system can evolve into a real-time, intelligent, and scalable inventory management solution, capable of reducing wastage and improving efficiency across diverse retail and supply chain environments.

APPENDICES

APPENDIX 1

CODING

Perishable Inventory Risk Engine

Backend (Flask -app.py):
```
from flask import Flask, request, jsonify

app = Flask(__name__)

@app.route("/predict", methods=["POST"])
def predict():
    data = request.json

    qty = float(data["quantity"])
    rate = float(data["consumption_rate"])
    days = float(data["expiry_days"])

    # simple risk logic (replace with ML model later)
    risk = (rate * days) / (qty + 1)

    status = "Low"
    if risk > 5:
        status = "High"
    elif risk > 2:
        status = "Medium"

    return jsonify({"risk": risk, "status": status})

if __name__ == "__main__":
    app.run(debug=True)
```

Frontend (HTML – index.html):
```
<!DOCTYPE html>
<html>
<body>

<h2>Inventory Risk Checker</h2>

<input id="q" placeholder="Quantity"><br><br>
<input id="r" placeholder="Consumption Rate"><br><br>
```

```
<input id="d" placeholder="Expiry Days"><br><br>

<button onclick="predict()">Predict</button>

<p id="result"></p>

<script>
function predict(){
   fetch("/predict", {
      method:"POST",
      headers:{"Content-Type":"application/json"},
      body:JSON.stringify({
         quantity: document.getElementById("q").value,
         consumption_rate: document.getElementById("r").value,
         expiry_days: document.getElementById("d").value
      })
   })
   .then(res => res.json())
   .then(data => {
      document.getElementById("result").innerHTML =
      "Risk: " + data.risk + " | Status: " + data.status;
   });
}
</script>

</body>
</html>
```

APPENDIX 2

SCREENSHOTS

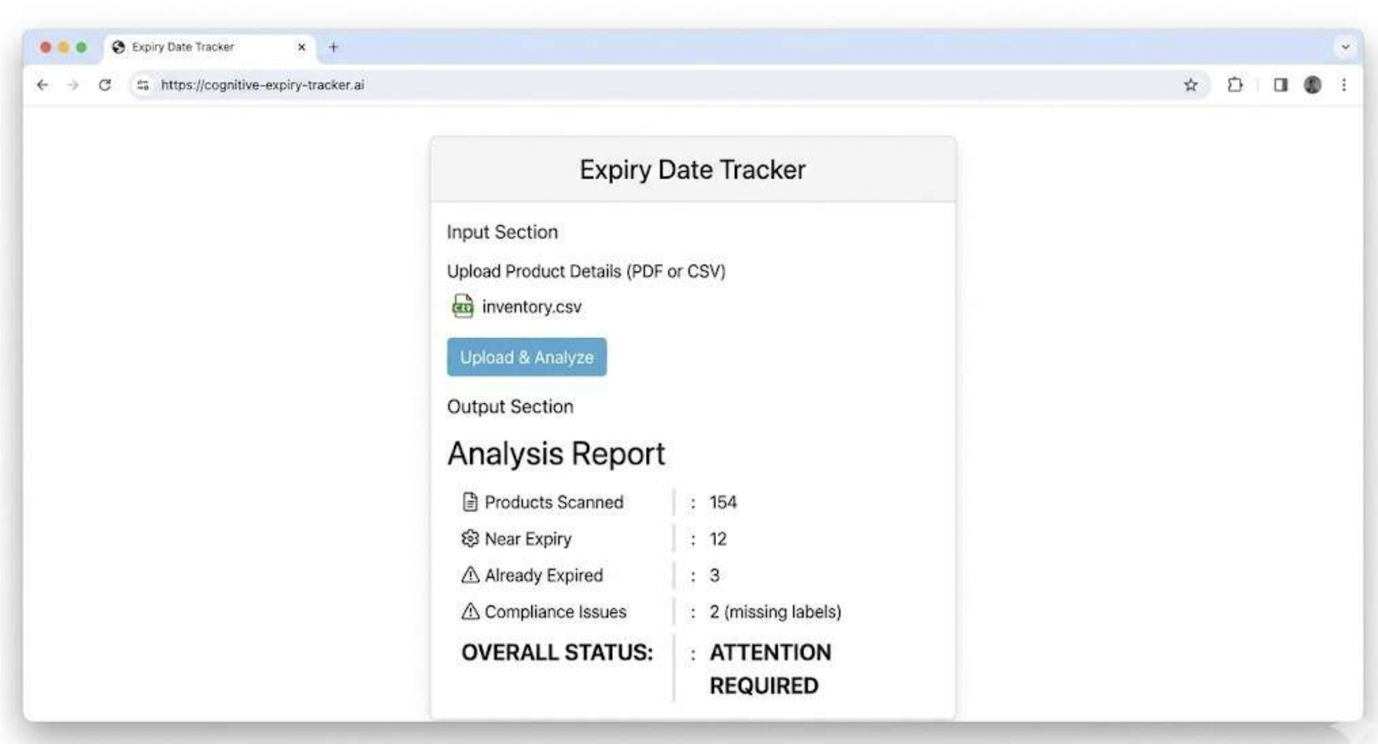


Fig A2.1 UI of the Web Interface for Perishable Inventory Risk Engine

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